

AAYUSH KUMAR

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EDUCATION

Indian Institute of Information Technology Dharwad

B.Tech - Electronics and Communication Engineering **CGPA: 7.87**

COURSEWORK / SKILLS

- Data Structures & Algorithms
- Operating Systems
- Database Management
- Artificial Intelligence
- Software Defined Networks
- React Native
- OOPS Concept
- Mobile App Development
- Computer Networks

PROJECTS

AlzVision-AI

- AI-powered mobile application for Alzheimer MRI analysis using hybrid CNN-Transformer models with real-time backend inference.
- Developed cross-platform React Native interface for MRI upload, prediction visualization, and confidence scoring.
- Integrated FastAPI backend delivering real-time predictions with responsive UI design.

Intelligent Perception Assistant for Visually Impaired

- AI-powered assistant providing real-time environmental understanding through multimodal perception and NLP.
- Implemented contextual reasoning using speech interaction and AI models.
- Integrated APIs including calendar, contacts, search, and location services.

Pulse5G

- 5G edge-based real-time incident detection and alerting system.
- Focused on low-latency edge computing architecture for fast analysis.

Leaf Disease Detection (LeafLens)

- Developed plant disease detection model using CNN and Random Forest.

RESEARCH

GEN-AI RAG Predictive Semantic Caching for 5G

Ongoing Research

- Developing a predictive semantic caching framework for 5G edge/MEC networks using GenAI-RAG-based demand forecasting.
- Implemented RMP-SC, TSK-SC, and LDP-SC caching strategies with mobility-aware user demand and limited edge-cache capacity.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, JavaScript, TypeScript, Kotlin

Machine Learning: Scikit-Learn, TensorFlow, CNNs, Transformers, Deep Learning, NLP

Databases: MySQL, MongoDB, Firebase Firestore, Room Database

Frameworks & Technologies: FastAPI, Rest API, React Native, Docker, Kubernetes, Helm, AWS EC2

Developer Tools: Git, GitHub, Linux, VS Code, Android Studio, IntelliJ IDEA

EXTRACURRICULAR

Founder - brancovenn.com

- Built an independent platform for experimenting with software ideas and product development.

Sim Gamepad (Ongoing Startup)

- Implemented a virtual gamepad system using TCP/UDP socket programming to transmit mobile inputs as controller signals, focusing on low-latency networking and real-time data handling.

Open Source Contributor (GitHub)

- Contributed to major open-source projects by submitting pull requests: [AWS Load Balancer Controller](#), [SigNoz](#), [Rocket.Chat](#), [Zulip](#), [Appwrite](#).
- Implemented bug fixes and feature enhancements while collaborating with maintainers through code reviews and improving code quality across repositories.